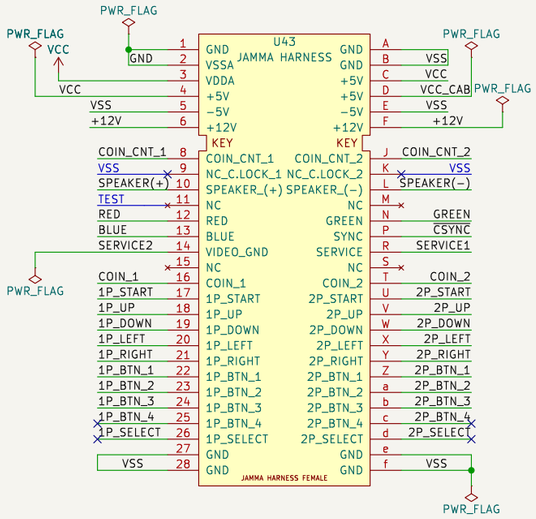
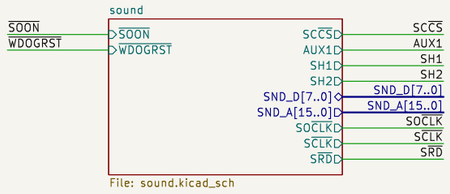
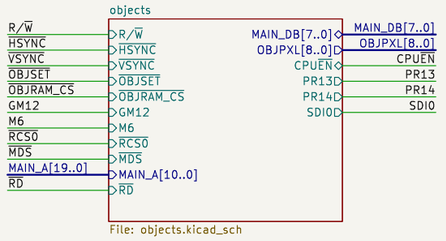
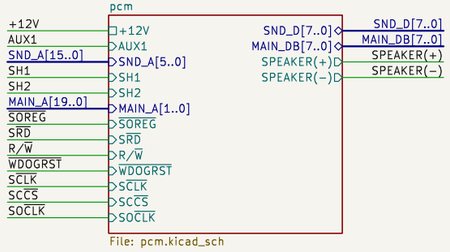
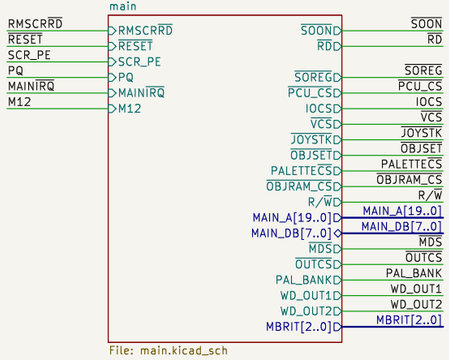
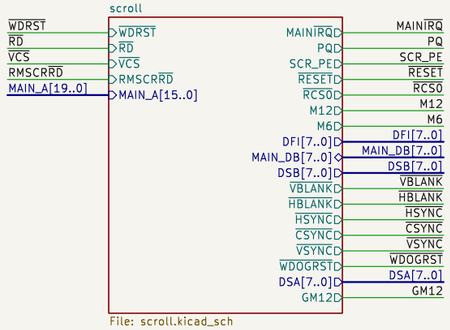


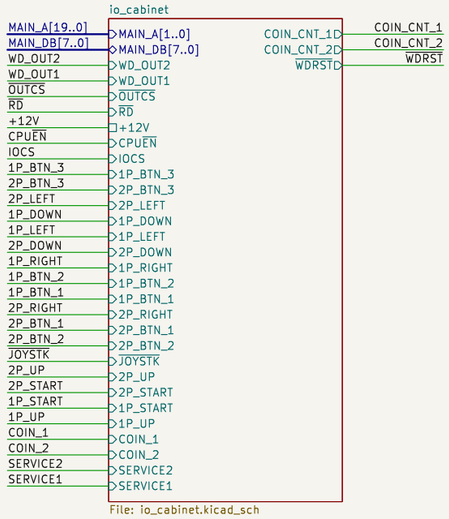


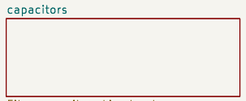


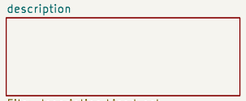


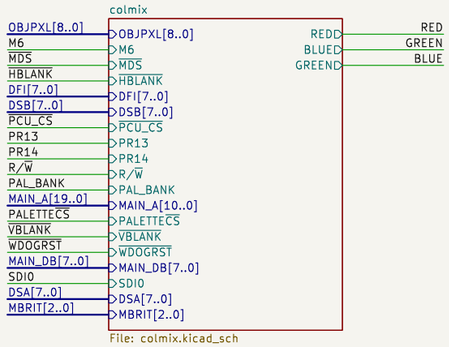










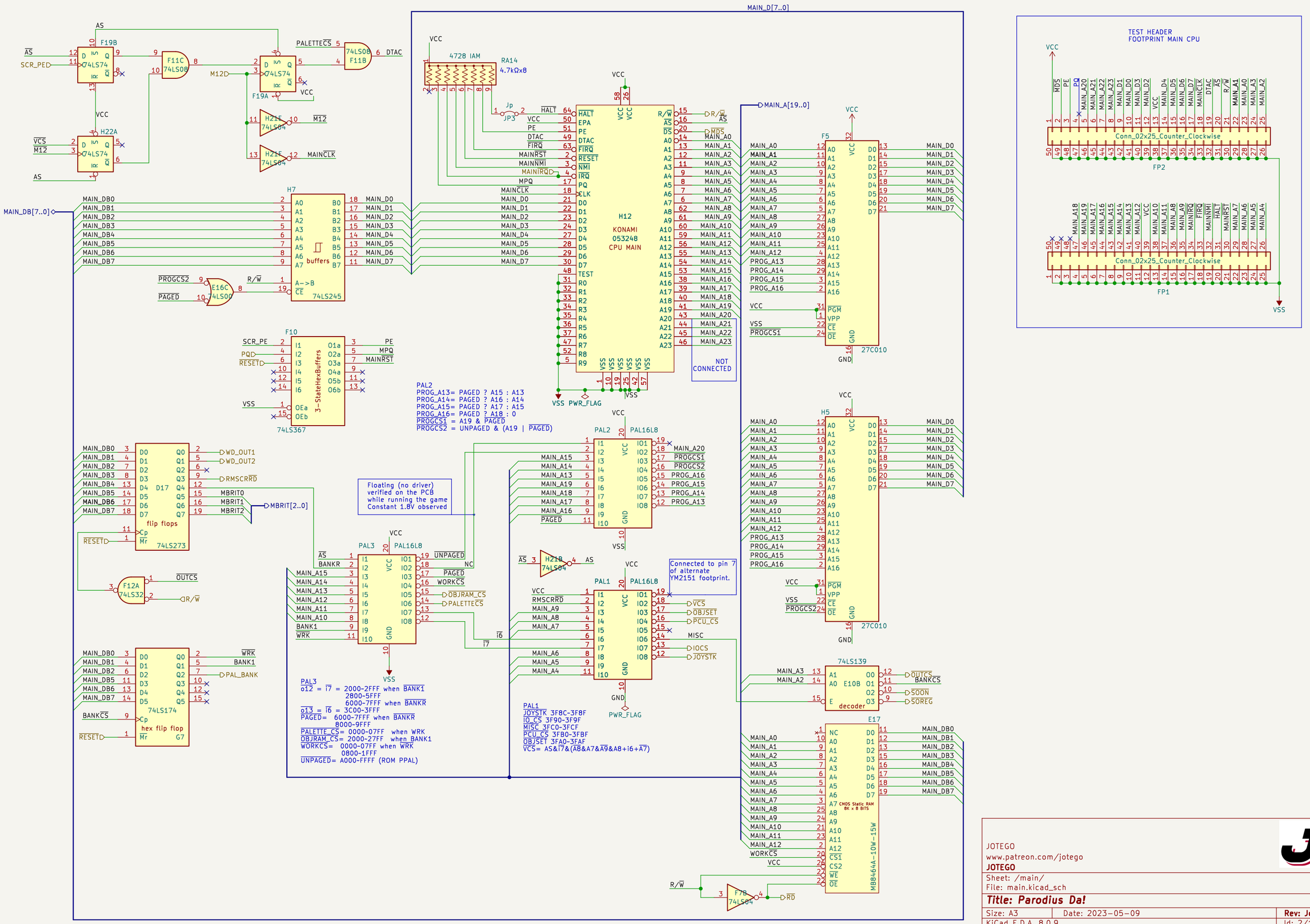


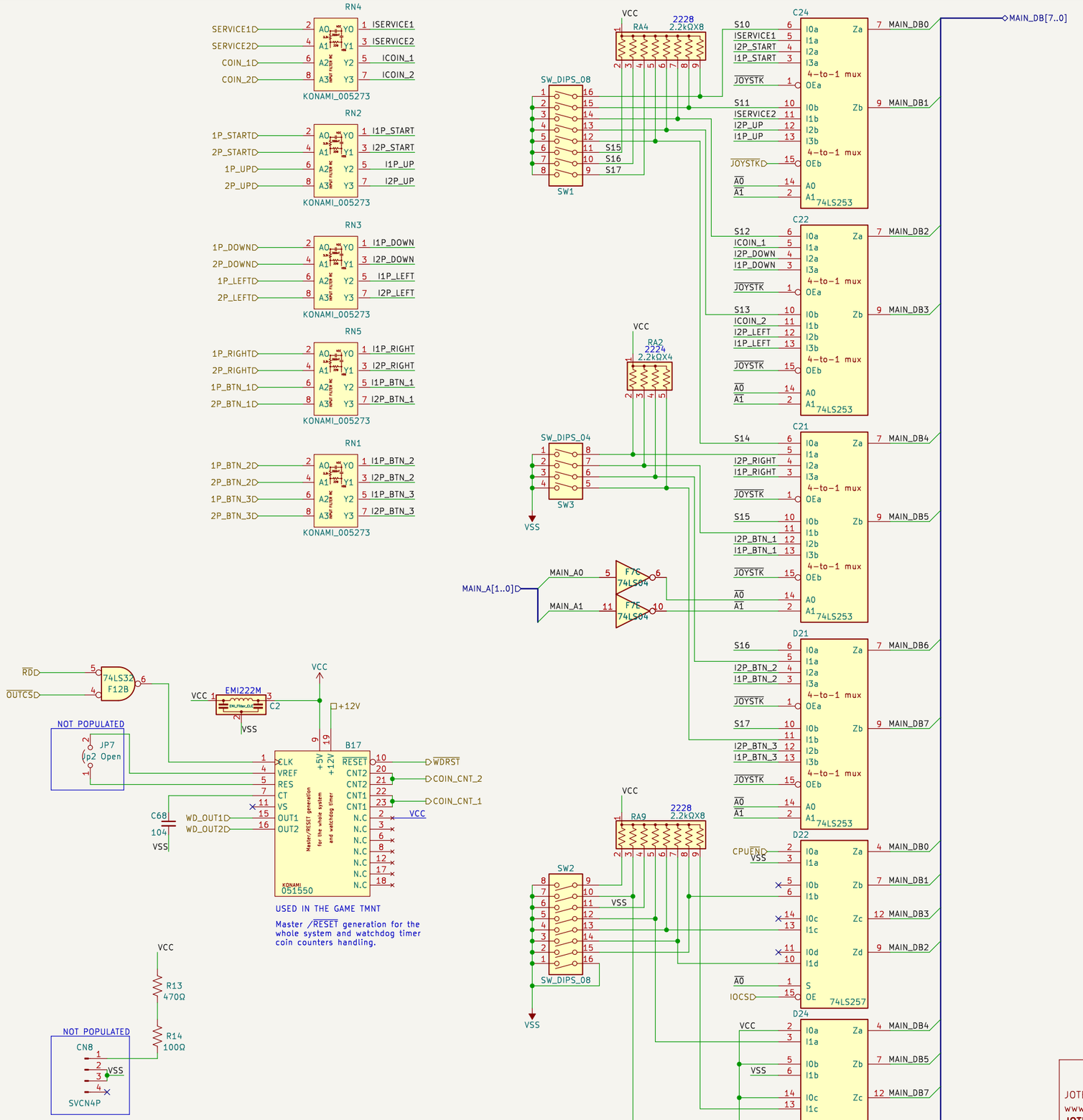
JOTEGO
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Sheet: /
File: parodius.kicad_sch

Title: Parodius Da!

Size: A3	Date: 2023-05-09	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.9		Id: 1/11


MAIN_D[7..0]
MAIN_A[19..0]
MAIN_DB[7..0]
TEST HEADER
FOOTPRINT MAIN CPU
Conn_02x25_Counter_Clockwise
FP1
FP2
H12
KONAMI
053248
CPU MAIN
H7
buffers
74LS245
F5
H5
27C010
F19B
F19A
F11C
F11B
H22A
H21E
H21B
F12A
F10
74LS367
3-StateHexBuffers
flip flops
74LS273
hex flip flop
74LS174
74LS139
decoder
E17
CMOS Static RAM
8K x 8 BITS
MB8464A-10W-15W
RA14
4.7kΩx8
JP3
PAL1
PAL2
PAL3
PAL16L8
PWR_FLAG
NOT
CONNECTED
Floating (no driver)
verified on the PCB
while running the game
Constant 1.8V observed
Connected to pin 7
of alternate
YM2151 footprint.
PAL2
PROG_A13= PAGED ? A15 : A13
PROG_A14= PAGED ? A16 : A14
PROG_A15= PAGED ? A17 : A15
PROG_A16= PAGED ? A18 : 0
PROGCS1 = A19 & PAGED
PROGCS2 = UNPAGED & (A19 | PAGED)
PAL3
o12 = i7 = 2000-2FFF when BANK1
2800-5FFF
6000-7FFF when BANKR
o13 = i6 = 3C00-3FFF
PAGED= 6000-7FFF when BANKR
8000-9FFF
PALETTE_CS= 0000-07FF when WRK
OBJRAM_CS= 2000-27FF when BANK1
WORKCS= 0000-07FF when WRK
0800-1FFF
UNPAGED= A000-FFFF (ROM PPAL)
PAL1
JOYSTK 3F8C-3F8F
IO_CS 3F90-3F9F
MISC 3FC0-3FCF
PCU_CS 3FB0-3FBF
OBJSET 3FA0-3FAF
VCS= AS&i7&(A8&A7&A9&AB+i6+A7)
JOTEGO
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JOTEGO
Sheet: /main/
File: main.kicad_sch
Title: Parodius Da!
Size: A3
Date: 2023-05-09
KiCad E.D.A. 8.0.9
Rev: José Tejada G.
Id: 2/11


RN4
SERVICE1
SERVICE2
COIN_1
COIN_2
ISERVICE1
ISERVICE2
ICOIN_1
ICOIN_2
KONAMI_005273
RN2
1P_START
2P_START
1P_UP
2P_UP
I1P_START
I2P_START
I1P_UP
I2P_UP
KONAMI_005273
RN3
1P_DOWN
2P_DOWN
1P_LEFT
2P_LEFT
I1P_DOWN
I2P_DOWN
I1P_LEFT
I2P_LEFT
KONAMI_005273
RN5
1P_RIGHT
2P_RIGHT
1P_BTN_1
2P_BTN_1
I1P_RIGHT
I2P_RIGHT
I1P_BTN_1
I2P_BTN_1
KONAMI_005273
RN1
1P_BTN_2
2P_BTN_2
1P_BTN_3
2P_BTN_3
I1P_BTN_2
I2P_BTN_2
I1P_BTN_3
I2P_BTN_3
KONAMI_005273
VCC
RA4
2228
2.2kΩX8
SW_DIPS_08
SW1
S15
S16
S17
C24
S10
ISERVICE1
I2P_START
I1P_START
JOYSTK
S11
ISERVICE2
I2P_UP
I1P_UP
A0
A1
74LS253
MAIN_DB0
MAIN_DB1
C22
S12
ICOIN_1
I2P_DOWN
I1P_DOWN
S13
ICOIN_2
I2P_LEFT
I1P_LEFT
MAIN_DB2
MAIN_DB3
RA2
2224
2.2kΩX4
SW_DIPS_04
SW3
VSS
C21
S14
I2P_RIGHT
I1P_RIGHT
S15
I2P_BTN_1
I1P_BTN_1
MAIN_DB4
MAIN_DB5
MAIN_A[1..0]
MAIN_A0
MAIN_A1
74LS04
D21
S16
I2P_BTN_2
I1P_BTN_2
S17
I2P_BTN_3
I1P_BTN_3
MAIN_DB6
MAIN_DB7
MAIN_DB[7..0]
4-to-1 mux
RD
OUTCS
74LS32
F12B
NOT POPULATED
JP7
Jp2 Open
EMI222M
C2
+12V
B17
CLK
VREF
RES
CT
VS
OUT1
OUT2
RESET
CNT2
CNT2
CNT1
CNT1
N.C
WDRST
COIN_CNT_2
COIN_CNT_1
Master /RESET generation for the whole system and watchdog timer
KONAMI
051550
C68
104
WD_OUT1
WD_OUT2
USED IN THE GAME TMNT
Master /RESET generation for the whole system and watchdog timer coin counters handling.
R13
470Ω
R14
100Ω
CN8
SVCN4P
SW2
SW_DIPS_08
RA9
2228
2.2kΩX8
CPUEN
IOCS
D22
D24
74LS257
JOTEGO
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JOTEGO
Sheet: /io_cabinet/
File: io_cabinet.kicad_sch
Title: Parodius Da!
Size: A3
Date: 2023-05-09
Rev: José Tejada G.
KiCad E.D.A. 8.0.9
Id: 3/11

NOT POPULATED

RA8 4kX8 · RA13 4kX8 · RA12 4kX8 · RA10 4kX8

GFXROMD0 GFXROMD1 GFXROMD2 GFXROMD3 GFXROMD4 GFXROMD5 GFXROMD6 GFXROMD7
GFXROMD8 GFXROMD9 GFXROMD10 GFXROMD11 GFXROMD12 GFXROMD13 GFXROMD14 GFXROMD15
GFXROMD16 GFXROMD17 GFXROMD18 GFXROMD19 GFXROMD20 GFXROMD21 GFXROMD22 GFXROMD23
GFXROMD24 GFXROMD25 GFXROMD26 GFXROMD27 GFXROMD28 GFXROMD29 GFXROMD30 GFXROMD31

GFXROMA[17..0]

CN7 — Connector_Generic_Female_First

K13 / K19 — 955D05-06-07-08 — KONAMI — MASK 4M - 16BIT — HN62414P — CMOS ROM

J15 — KONAMI 051962 TILE DRAWING

DSA[7..0] SCRA

DFI[7..0] FIX

VBLANK HBLANK VSYNC HSYNC CSYNC

C15C 74LS07 · R18 470Ω · R20 100Ω · CSYNC

MAIN_DB[7..0] · DSB[7..0] SCRB

G22 XTAL_OSC 24MHz · CP1 104 · 74LS04 H21A · 74LS04 H21C

H22B 74LS74 · WDRST · WDOGRST

TEST · JP5 Jp_2_Bridged · JP6 Jp_2_Bridged

SCRAMA[12..0] · VD[15..0]

H24 · J24 — MB8464A-10W-15W — CMOS Static RAM 8K x 8 BITS

MAIN_A[15..0]

J20 KONAMI 052109 TILE MAP GENERATOR

LAYER A SCROLL X · LAYER B SCROLL X · TILE ATTRIBUTES · GFX ROM ADDRESS · VRAM ADDRESS · GFX ROM read · VRAM DATA · CPU DATA

MAIN_DB[7..0]

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Sheet: /scroll/
File: scroll.kicad_sch

Title: Parodius Da!

Size: A3 | Date: 2023-05-09 | Rev: José Tejada G.
KiCad E.D.A. 8.0.9 | Id: 4/11

NOT POPULATED

MAIN_A[10..0]

OBJROMA[18..0]

OBJROMD[31..0]

OBJROMD[7..0]

ORAMA[10..0]

MAIN_DB[7..0]

PL[6..0]

HP[9..0]

BR[2..0]

AR[2..0]

OBJPXL[8..0]

VCC

VSS

RA1 4kX8

RA11 4kX8

RA5 4kX8

RA3 4kX8

OBJROMD0 OBJROMD1 OBJROMD2 OBJROMD3 OBJROMD4 OBJROMD5 OBJROMD6 OBJROMD7

OBJROMD8 OBJROMD9 OBJROMD104 OBJROMD115 OBJROMD126 OBJROMD137 OBJROMD148 OBJROMD159

OBJROMD162 OBJROMD173 OBJROMD184 OBJROMD195 OBJROMD206 OBJROMD217 OBJROMD228 OBJROMD239

OBJROMD242 OBJROMD253 OBJROMD264 OBJROMD275 OBJROMD286 OBJROMD297 OBJROMD308 OBJROMD319

CN6

Conn_02x32_Row_Letter_First

a1 a2 a3 a4 a5 a6 a7 a8 a9 a10 a11 a12 a13 a14 a15 a16 a17 a18 a19 a20 a21 a22 a23 a24 a25 a26 a27 a28 a29 a30 a31 a32

b1 b2 b3 b4 b5 b6 b7 b8 b9 b10 b11 b12 b13 b14 b15 b16 b17 b18 b19 b20 b21 b22 b23 b24 b25 b26 b27 b28 b29 b30 b31 b32

OBJROMA0 OBJROMA1 OBJROMA2 OBJROMA3 OBJROMA4 OBJROMA5 OBJROMA6 OBJROMA7 OBJROMA8 OBJROMA9 OBJROMA10 OBJROMA11 OBJROMA12 OBJROMA13 OBJROMA14 OBJROMA15 OBJROMA16 OBJROMA17 OBJROMA18

OBJROMD0 OBJROMD1 OBJROMD2 OBJROMD3 OBJROMD4 OBJROMD5 OBJROMD6 OBJROMD7 OBJROMD8 OBJROMD9 OBJROMD10 OBJROMD11 OBJROMD12 OBJROMD13 OBJROMD14 OBJROMD15

OBJROMD15 OBJROMD16 OBJROMD17 OBJROMD18 OBJROMD19 OBJROMD20 OBJROMD21 OBJROMD22 OBJROMD23 OBJROMD24 OBJROMD25 OBJROMD26 OBJROMD27 OBJROMD28 OBJROMD29 OBJROMD30 OBJROMD31

U41
KONAMI
053245
OBJECT GENERATOR

OBJECT ROM ADDRESS BUS

MAIN ADDRESS BUS

OBJECT RAM ADDRESS BUS

OBJECT RAM DATA BUS

GENERATOR TO DRAWING

U42
KONAMI
053244
OBJECT DRAWING

MAIN

PRIORITY CONTROL UNIT

GENERATOR TO DRAWING

HM6116P-3

J13
MCM2018

2kB RAM

H14
buffers
A->B
74LS245

F12C
74LS32

R/W

OBJRAM_CS

F13D
74LS32

OBJRAM_RD

C14F
74LS04

OBJSET

F11D
74LS08

OBJRAM_CS

74LS32
F13C

CPUEN

RD

ROM's MASK OF THE SPRITES

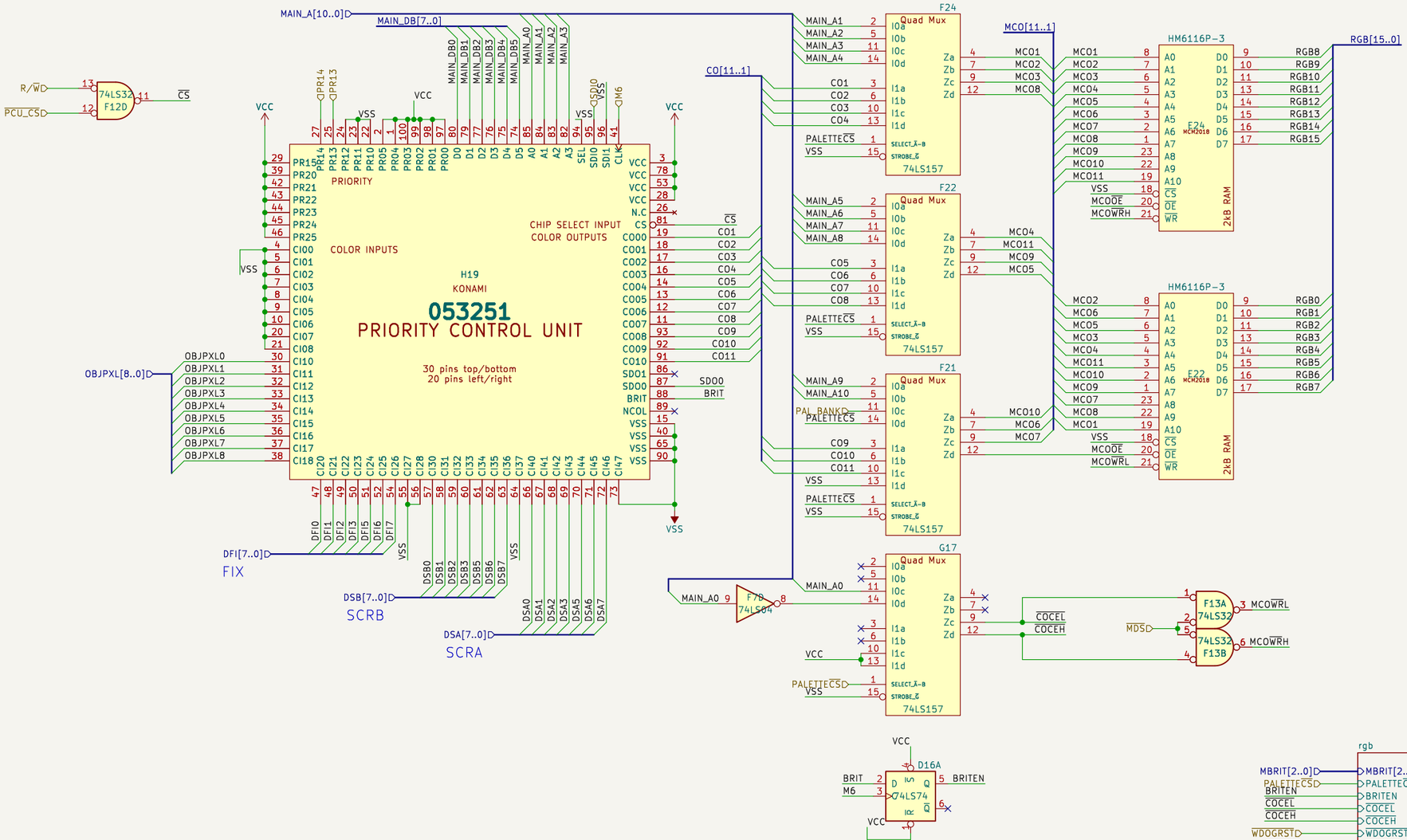
K2
K8
955D05-06-07-08
KONAMI
MASK 4M - 16BIT
CMOS ROM
HN62414P

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Sheet: /objects/
File: objects.kicad_sch

Title: Parodius Da!

Size: A3	Date: 2023-05-09	**Rev: José Tejada**
KiCad E.D.A. 8.0.9		Id: 5/11



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Sheet: /colmix/
File: colmix.kicad_sch

Title: Parodius Da!

Size: A3	Date: 2023-05-09	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.9		Id: 6/11

RGB[15..0]

MAIN_DB[7..0]

E21 — 74LS245 buffers (A->B, CE)

MAIN_DB0 2
MAIN_DB1 3
MAIN_DB2 4
MAIN_DB3 5
MAIN_DB4 6
MAIN_DB5 7
MAIN_DB6 8
MAIN_DB7 9

PALETTECS 1
COCEL 19

RGB0 18
RGB1 17
RGB2 16
RGB3 15
RGB4 14
RGB5 13
RGB6 12
RGB7 11

SDOOD 5 — C14C 74LS04 — 6 18

D19 — 74LS273 flip flops (Cp, Mr)

D0 3 — Q0 2 G3
D1 4 — Q1 5 G4
D2 7 — Q2 6 B0
D3 8 — Q3 9 B1
D4 13 — Q4 12 B2
D5 14 — Q5 15 B3
D6 17 — Q6 16 B4
D7 18 — Q7 19 SHADOW

M6 11
VCC 1

E20 — 74LS245 buffers (A->B, CE)

MAIN_DB0 2
MAIN_DB1 3
MAIN_DB2 4
MAIN_DB3 5
MAIN_DB4 6
MAIN_DB5 7
MAIN_DB6 8
MAIN_DB7 9

PALETTECS 1
COCEH 19

RGB8 18
RGB9 17
RGB10 16
RGB11 15
RGB12 14
RGB13 13
RGB14 12
RGB15 11

D20 — 74LS273 flip flops (Cp, Mr)

D0 3 — Q0 2 R0
D1 4 — Q1 5 R1
D2 7 — Q2 6 R2
D3 8 — Q3 9 R3
D4 13 — Q4 12 R4
D5 14 — Q5 15 G0
D6 17 — Q6 16 G1
D7 18 — Q7 19 G2

M6 11
VCC 1

F20 — 74LS164 High Speed 8-Bit Serial In Parallel-Out Shift Register

VBLANK 1
HBLANK 2
M6 8
WDOGRST 9

A, B → AND
CLK, MR → NOT
Q0 3, Q1 4, Q2 5, Q3 6, Q4 10, Q5 11, Q6 12, Q7 13

C15A 74LS07 1 2
C15B 74LS07 3 4

G4 3 — C17B 74LS07 — 4
G3 5 — C19C 74LS07 — 6
G2 13 — C20F 74LS07 — 12
G1 3 — C19B 74LS07 — 4
G0 9 — C17D 74LS07 — 8
13 — C15F 74LS07 — 12

B4 11 — C15E 74LS07 — 10
B3 9 — C15D 74LS07 — 8
B2 9 — C16D 74LS07 — 8
B1 11 — C16F 74LS07 — 10
B0 5 — C17C 74LS07 — 6

R4 11 — C20E 74LS07 — 10
R3 3 — C20B 74LS07 — 4
R2 5 — C20C 74LS07 — 6
R1 1 — C20A 74LS07 — 2
R0 9 — C20D 74LS07 — 8

USED IN THE GAME LETHAL ENFORCERS

S2 — KONAMI 052535 5-bit Video DAC (DIGITAL INPUT, ANALOG OUTPUT RGB)

VCC
D4 (Pull Up*), D3*, D2*, D1*, D0*, BK*
AO 8 → GREEN
VSS

SHADOW 3 — C16B 74LS07 — 4 — R21 N.M
MBRIT2 13 — C17F 74LS07 — 12 — R16 N.M

MBRIT1 13 — C19F 74LS07 — 12 — R25 N.M
MBRIT0 11 — E19E 74LS07 — 10 — R15 N.M

S1 — KONAMI 052535 5-bit Video DAC (DIGITAL INPUT, ANALOG OUTPUT RGB)

VCC
D4 (Pull Up*), D3*, D2*, D1*, D0*, BK*
AO 8 → BLUE

SHADOW 13 — C16E 74LS07 — 12 — R23 N.M
MBRIT2 11 — C17E 74LS07 — 10 — R26 N.M

MBRIT1 13 — 74LS00 E16D — 11 — MBRIT1 — C19E 74LS07 — 10 — R24 N.M
MBRIT0 5 — 74LS00 E16B — 6 — MBRIT0 — E19A 74LS07 — 2 — R22 118 Ohm
BRITEN
MBRIT2 2 — 74LS00 E16A — 3 — MBRIT2
MBRIT[2..0]

S3 — KONAMI 052535 5-bit Video DAC (DIGITAL INPUT, ANALOG OUTPUT RGB)

VCC
D4 (Pull Up*), D3*, D2*, D1*, D0*, BK*
AO 8 → RED

SHADOW 1 — C16A 74LS07 — 2 — R12 N.M
MBRIT2 1 — C17A 74LS07 — 2 — R27 N.M

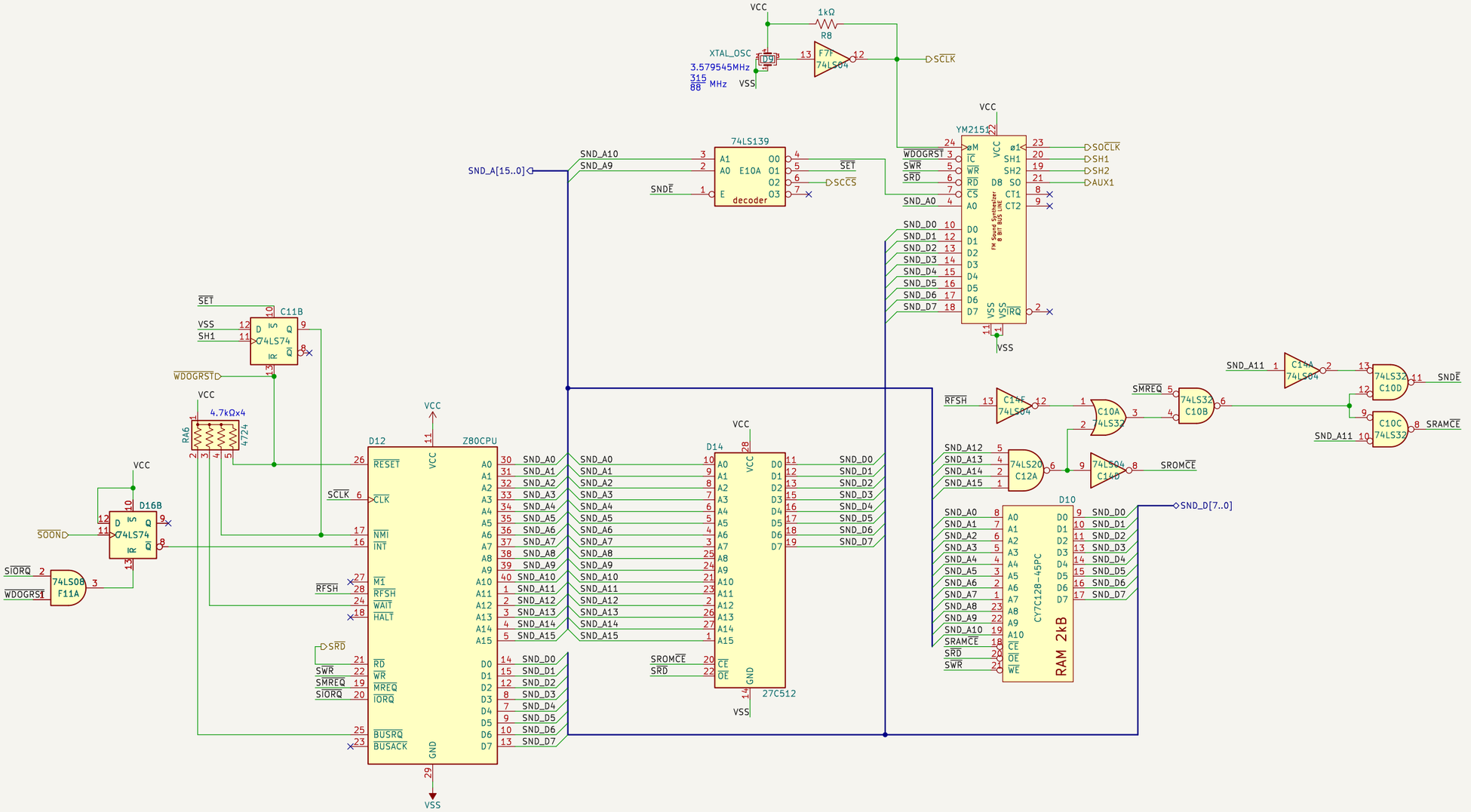
MBRIT1 1 — C19A 74LS07 — 2 — R19 N.M
MBRIT0 13 — E19F 74LS07 — 12 — R17 N.M

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Sheet: /colmix/rgb/
File: rgb.kicad_sch

Title: Parodius Da!

Size: A3	Date: 2023-05-09	**Rev: José Tejada G.**
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Sheet: /sound/
File: sound.kicad_sch

Title: Parodius Da!

Size: A3	Date: 2023-05-09	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.9		Id: 8/11

NOT POPULATED

RA[19..0]

Communication between the main CPU
and the audio CPU.

RD[7..0]

Connected to pin 21 of alternate YM2151 footprint.

JP_2_Bridged

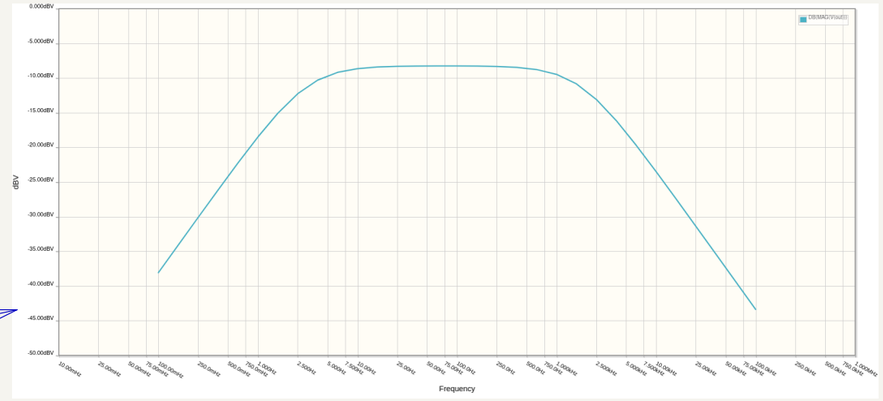
C7 KONAMI 053260 FOUR VOICE PCM/ADPCM SOUND CHIP FOUR 8-BIT PORTS Pulse Code Modulation

C5 KONAMI 955D04 MASK 4M – 8BIT 62414

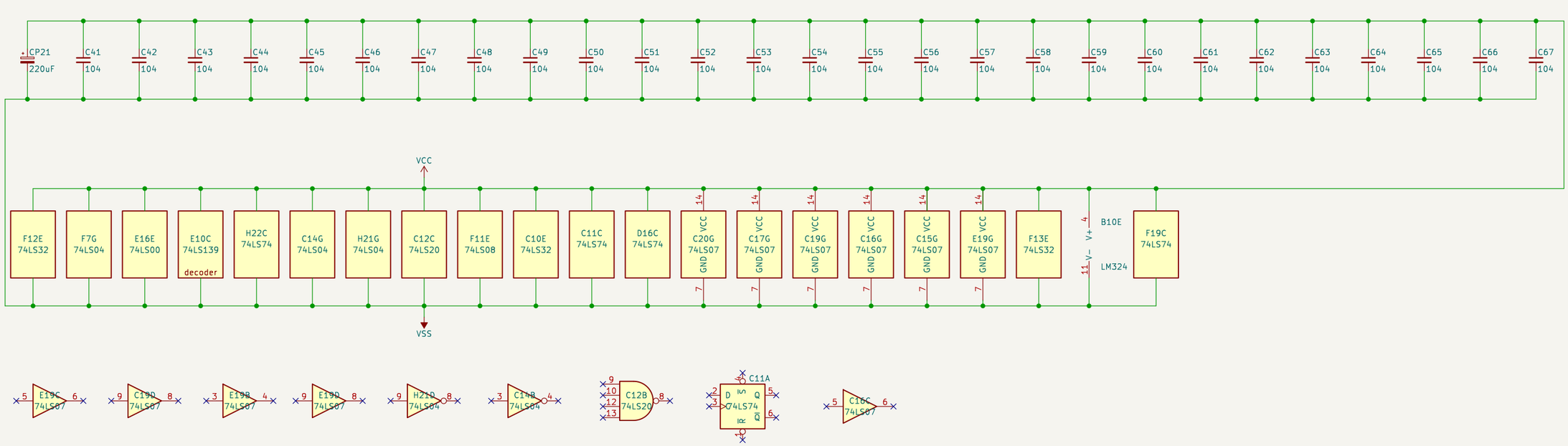
YM3012 2-CHANNEL SERIAL INPUT FLOATING D/A CONVERTER (DAC-HG)

MB3722

LM324



JOTEGO www.patreon.com/jotego **JOTEGO**		
Sheet: /pcm/ File: pcm.kicad_sch		
Title: Parodius Da!		
Size: A3	Date: 2023-05-09	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.9		Id: 9/11



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Sheet: /capacitors/
File: capacitors.kicad_sch

Title: Parodius Da!

Size: A3	Date: 2023-05-09	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.9		Id: 10/11

Parodius / Parodius DA!
1990

Parodius! From Myth to Laughter, released in Japan as Parodius Da!
Shinwa kara Owarai e[a] and outside Japan as Parodius
(from "Parody Gradius"), is a shoot 'em up arcade video game and the
second title in the Parodius series produced by Konami.
The European SNES version is also known as Parodius: Non-Sense Fantasy.
The gameplay is stylistically very similar to the Gradius series, but
the graphics and music are intentionally absurd.
As a result of its localised title, From Myth to Laughter is often
mistaken as the original game of the series. The lesser known original game,
Parodius: The Octopus Saves the Earth, was released for the MSX computer in
Japan.

HARDWARE DESCRIPTION

Video Hardware : 052109 051962 053245 053244 053251
CPU : 053248
KNOWN KONAMI TMNT2 BASED GAME LIST
Detana!! Twin Bee / Bells & Whistles / Here Comes Twinbee!!
Golfing Greats
Lightning Fighters / Trigon
Parodius / Parodius DA!
Premier Soccer
Quiz Gakumon no Susume
Rollergames
Sunset Riders
Surprise Attack
Teenage Mutant Ninja Turtles : Turtles in Time

Sequels : Gokujyou Parodius / Fantastic Journey (1994)
Sexy Parodius (1996)

Source: System 16 The Arcade Museum
wikipedia.org/wiki/Parodius_Da!

JOTEGO
www.patreon.com/jotego
JOTEGO
Sheet: /description/
File: description.kicad_sch
Title: Parodius Da!
Size: A3 | Date: 2023-05-09 | **Rev: José Tejada G.**
KiCad E.D.A. 8.0.9 | Id: 11/11